ShopLite orders platform

Consumer lag on the orders queue after worker restarts

Detailed root cause analysis

ShopLite Commerce

Summary

Field	Value
RCA ID	RCA-2026-011
Date	2026-04-10
Short description	A memory leak in orders-worker 2.7 made workers restart every 20 to 30 minutes. Each restart forced a consumer group rebalance, and lag on the orders topic grew to 18,000 messages.
Authors	Jonas Weber
Stakeholders	Jonas Weber, Mei Chen, SRE on-call
Status	Final
Issue environment	GCP, production, europe-west1, GKE cluster shop-prod
Incident start time	2026-04-08 12:14 UTC
Incident duration	48 minutes
Affected services or applications	orders-worker, order-queue
Severity level	Medium
Trigger	Alert OrderQueueConsumerLagHigh
Related SOPs	SOP-QUEUE-002

Problem statement

Consumer lag on the orders topic grew to 18,000 messages. orders-worker pods were killed with "OOMKilled (exit code 137)" every 20 to 30 minutes. Scaling workers did not help, because the new workers also ran out of memory and every restart paused all consumers.

Impact

Description of impact

External customers in all regions. Order confirmations were late for two hours.

Key metrics affected

Confirmations up to 15 minutes late for about 3,000 orders. No orders lost. No SLA breach.

Customer feedback

18 support tickets. No escalations.

Immediate resolution

Rolled orders-worker back from 2.7 to 2.6.

Event timeline

Timestamp	Event description	Action taken	Responsibility
2026-04-08 10:30	orders-worker 2.7 released.	None.	Orders team
2026-04-08 11:05	First worker restart with OOMKilled (exit code 137).	None. There was no alert on worker restarts.	SRE on-call
2026-04-08 12:14	Alert OrderQueueConsumerLagHigh fired with lag at 12,500.	On-call followed SOP-QUEUE-002 and scaled workers to 4.	SRE on-call
2026-04-08 12:41	Lag still growing. On-call saw the restart count.	Rolled orders-worker back to 2.6.	SRE on-call
2026-04-08 13:02	Lag back under 1,000.	Closed the incident.	SRE on-call

Analysis

5-whys analysis

1st why

- Question: Why did consumer lag grow?
- Answer: Workers kept restarting, and each restart paused all consumers for a rebalance of 20 to 40 seconds.

2nd why

- Question: Why did workers restart?
- Answer: They ran out of memory and were killed by Kubernetes.

3rd why

- Question: Why did they run out of memory?
- Answer: Version 2.7 cached every processed order in memory and never cleared the cache.

4th why

- Question: Why was the leak not caught before release?
- Answer: The release pipeline has no long running memory test.

5th why

- Question: Why did on-call scale instead of rolling back?
- Answer: The lag procedure did not say to check worker memory or restarts before scaling.

Fishbone analysis

Process

- The lag procedure did not check worker health before scaling.

Tools

- No memory soak test in the worker release pipeline.

Data

- No alert on worker restarts.

Root causes

Based on the analysis above, the root causes of the incident are:

1. orders-worker 2.7 leaked memory through an order cache that was never cleared.
2. The lag procedure scaled workers without first checking that they were healthy.

Action items

The following action items address the root causes above.

Action	Owner	Due date
Fix the order cache and release orders-worker 2.7.1.	Orders team	2026-04-10
Add a worker memory check to SOP-QUEUE-002 before scaling.	Mei Chen	2026-04-17
Alert when a worker restarts more than twice in 30 minutes.	SRE team	2026-04-17
Add a memory soak test to the worker release pipeline.	Orders team	2026-05-29